

TEM-1 β -lactamase variant-effect benchmark: PLM architectures and feature weights

Sequence-only prediction of β -lactamase functionality — a direction-finding benchmark

Summary

This benchmark tests whether protein-language-model (PLM) features predict TEM-1 β -lactamase functionality better than a model that sees only amino-acid identity, and compares three ESM-family architectures to identify which representation to build a final predictor on. Ground truth is the Firnberg 2014 deep-mutational-scanning (DMS) dataset: 4770 single-missense variants with a binary functional/non-functional label (13 wet-lab-panel variants held out and **not** scored here). All models are evaluated by out-of-fold (OOF) AUROC under 5-fold cross-validation across three splits — random, modulo-position, and **contiguous** (whole protein regions withheld) — with the contiguous split as the primary, honest test of generalization to unseen regions.

Classifier selection methodology. Every reported cell is a (feature_set, split) pair scored by four classifiers (LR, RF, XGBoost, SVM). The classifier reported for a feature set is fixed by its performance on the **modulo split**, never by the contiguous score being headlined — picking the classifier from the same number being reported would be circular and optimistically biased. All four classifiers' contiguous-split AUROC are also reported as a sensitivity block (§2) so the pre-specified pick is fully auditable. This selection rule was applied retroactively to a version of this benchmark that had picked the best-of-4 independently per split; the correction changed 3 of 21 cells by ≤ 0.007 AUROC, and did not change the identity of the best sequence-only cell.

Headline result. The best sequence-only PLM configuration (ESM C 600M, per-position surprisal + substitution one-hot, XGBoost — the pre-specified classifier for this feature set) reaches **contiguous AUROC 0.899**, versus the identity-only floor at **0.705** and a physicochemical floor at **0.726** (both floors: XGBoost, also pre-specified) — a gap of **+0.194 AUROC** that is overwhelmingly significant (DeLong $p \approx 3.6 \times 10^{-131}$; paired-bootstrap 95% CI [0.178, 0.210]). The premise of the project — that PLM features carry biological signal beyond position identity — holds on the hardest split, and the gap is, if anything, slightly larger under bias-free classifier selection than it first appeared.

This is a proof-of-concept benchmark. No final model is locked or saved; the sealed wet-lab holdout is untouched.

1. Thesis: PLM features beat the identity floor

The identity one-hot baseline encodes only *which position was mutated to which residue* — it can memorize position-level tolerance but knows no biology. Using the pre-specified classifier (XGBoost, chosen by modulo-split performance) it reaches contiguous AUROC **0.705**. Every PLM representation clears it by a wide margin, and the best PLM cell does so by **+0.194 AUROC**.

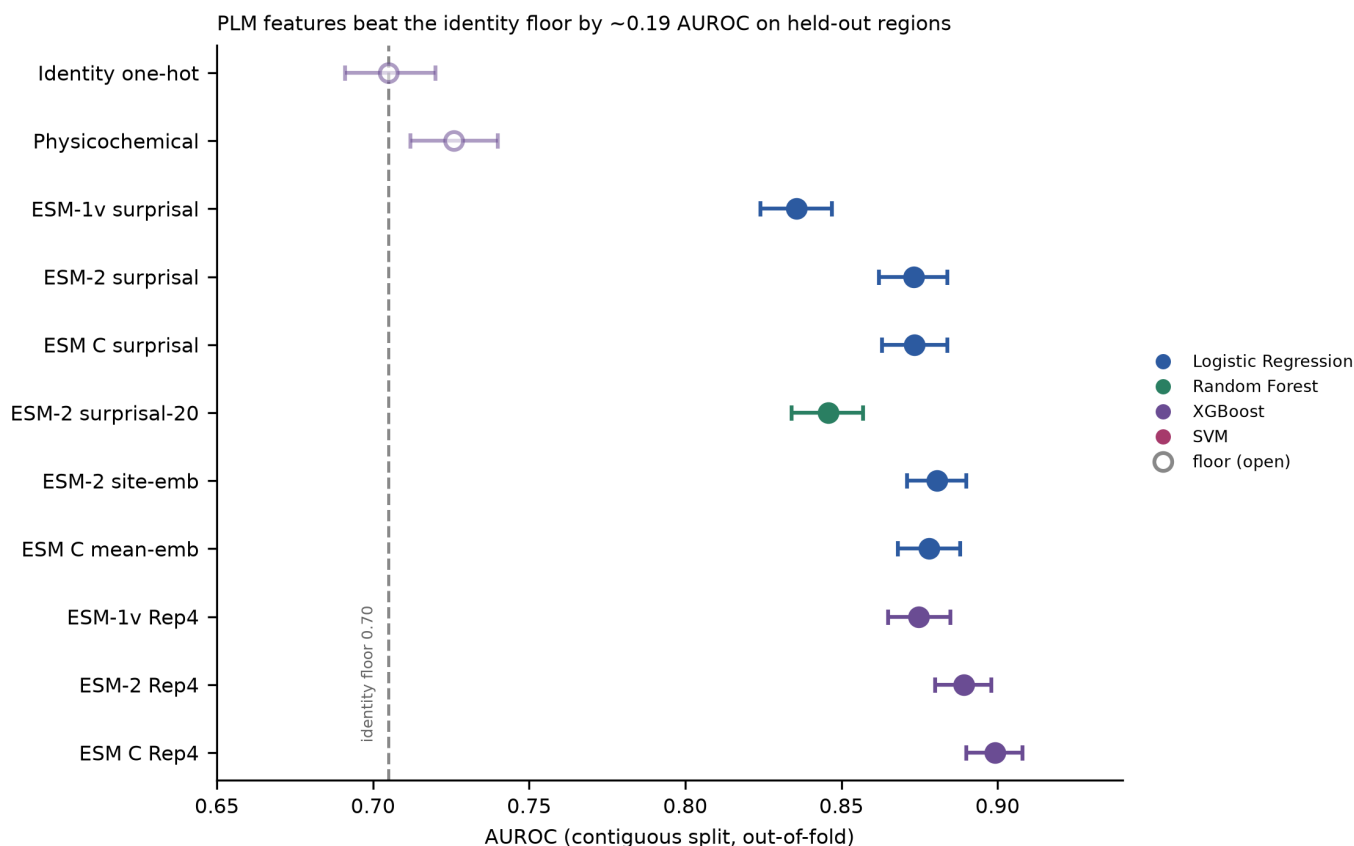


Figure 1. Contiguous-split OOF AUROC (pre-specified classifier per representation) with 95% bootstrap CIs. Open markers = floors (identity, physicochemical); filled = PLM. Marker colour = the pre-specified classifier (chosen by modulo-split AUROC, not by the contiguous score shown). Dashed line = identity floor (0.705). The ~0.19 gap between the floors and the PLM representations is the central result.

The significance test on the paired OOF predictions confirms the gap is not noise. Both classifiers below are the pre-specified (modulo-selected) choice for their feature set:

Comparison	PLM AUROC	Floor AUROC	Δ	DeLong p	Bootstrap 95% CI
ESM C Rep4 (XGBoost) vs Identity floor (XGBoost)	0.8991	0.7050	0.1942	3.65e-131	[0.1784,0.2096]

Note the floor classifier changed from Random Forest (the earlier, contiguous-biased pick) to XGBoost (the pre-specified pick) — the floor AUROC dropped slightly (0.7124 → 0.7050) as a result, which widens the reported gap rather than narrowing it.

2. Architecture comparison: ESM-2 vs ESM-1v vs ESM C

All three ESM-family architectures were scored on the same variants, splits, and classifiers. The comparison identifies which architecture and representation to carry forward. The classifier shown for each row is pre-specified per feature set (chosen by modulo-split AUROC); a full four-classifier sensitivity table follows for transparency.

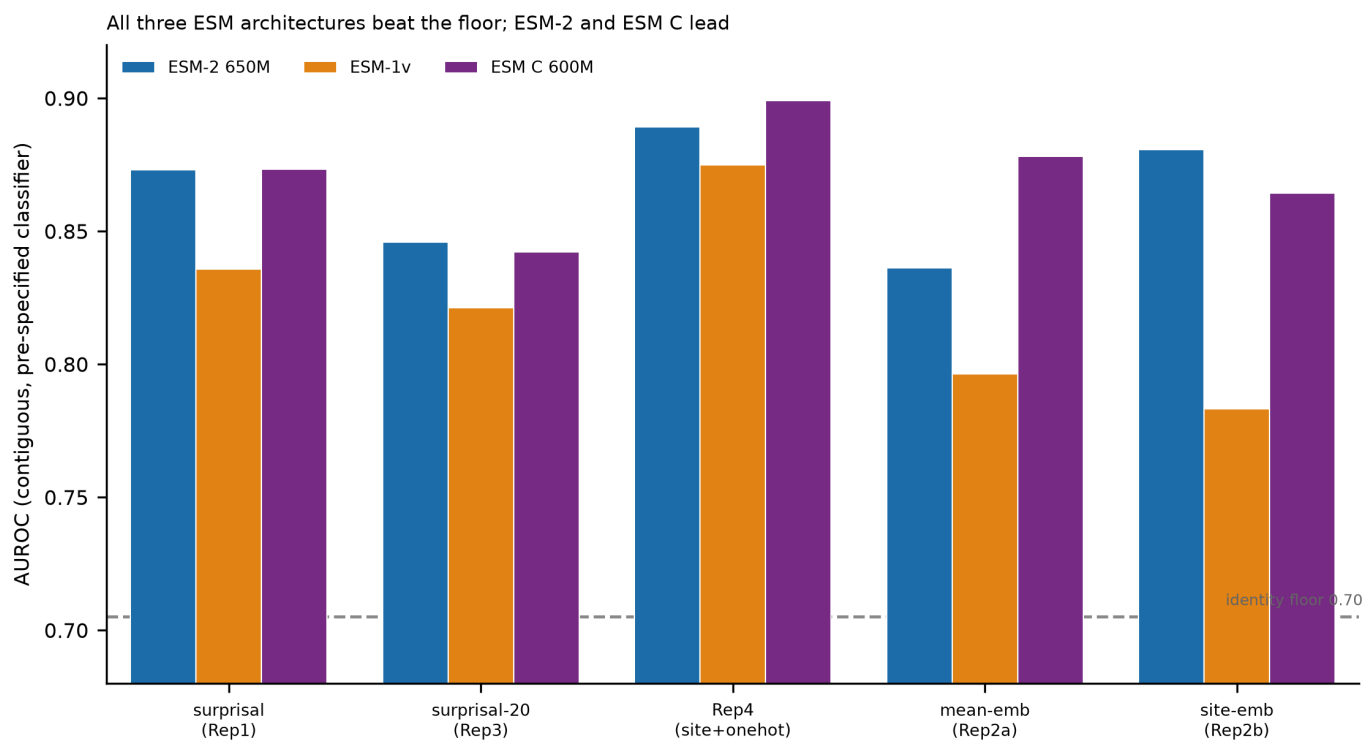


Figure 2. Contiguous AUROC (pre-specified classifier) by architecture and representation. All three architectures clear the floor across every representation.

Architecture	Representation	Contiguous AUROC	95% CI	Pre-specified classifier	Random	Modulo
Floor	Physicochemical	0.7258	[0.712,0.740]	XGBoost	0.7477	0.7287
Floor	Identity one-hot	0.7050	[0.691,0.720]	XGBoost	0.7382	0.7133
ESM-2 650M	Rep4 site+onehot	0.8892	[0.880,0.898]	XGBoost	0.9553	0.9420
ESM-2 650M	Rep1 surprisal + physchem	0.8835	[0.874,0.894]	Logistic Regression	0.8944	0.8881
ESM-2 650M	Rep2b site-emb	0.8807	[0.871,0.890]	Logistic Regression	0.9589	0.9445
ESM-2 650M	Rep1 surprisal	0.8731	[0.862,0.884]	Logistic Regression	0.8782	0.8770
ESM-2 650M	Rep2c Δ site-emb	0.8605	[0.850,0.871]	Logistic Regression	0.9513	0.9244
ESM-2 650M	Rep3 surprisal-20	0.8458	[0.834,0.857]	Random Forest	0.8950	0.8855
ESM-2 650M	Rep2a mean-emb	0.8363	[0.825,0.848]	Logistic Regression	0.9529	0.9261
ESM-1v	Rep4 site+onehot	0.8748	[0.865,0.885]	XGBoost	0.9527	0.9333
ESM-1v	Rep1 surprisal	0.8357	[0.824,0.847]	Logistic Regression	0.8459	0.8445
ESM-1v	Rep3 surprisal-20	0.8212	[0.809,0.833]	Random Forest	0.8945	0.8823
ESM-1v	Rep2c Δ site-emb	0.8206	[0.808,0.833]	Logistic Regression	0.9446	0.9152
ESM-1v	Rep2a mean-emb	0.7964	[0.783,0.810]	Logistic Regression	0.9454	0.9225
ESM-1v	Rep2b site-emb	0.7832	[0.771,0.797]	Logistic Regression	0.9567	0.9252
ESM C 600M	Rep4 site+onehot	0.8991	[0.890,0.908]	XGBoost	0.9575	0.9434
ESM C 600M	Rep2a mean-emb	0.8780	[0.868,0.888]	Logistic Regression	0.9588	0.9420
ESM C 600M	Rep1 surprisal	0.8733	[0.863,0.884]	Logistic Regression	0.8785	0.8777
ESM C 600M	Rep2b site-emb	0.8643	[0.854,0.874]	Logistic Regression	0.9596	0.9452

ESM C 600M	Rep2c Δ site-emb	0.8589	[0.848,0.869]	Logistic Regression	0.9533	0.9309
ESM C 600M	Rep3 surprisal-20	0.8422	[0.831,0.853]	Random Forest	0.8953	0.8842

Four-classifier sensitivity block (contiguous AUROC for all four classifiers on each feature set; **bold** = the pre-specified pick; -- = not run — RF/XGBoost were gated to ≤ 400 -dim feature sets and SVM to ≤ 200 -dim, so the 1,152–1,280-dim raw embedding representations (Rep2a/b/c) are LR-only by design):

Architecture	Representation	Pre-specified	LR	RF	XGB	SVM
Floor	Physicochemical	XGB	0.6082	0.7264	0.7258	0.6923
Floor	Identity one-hot	XGB	0.6797	0.7124	0.7050	0.6830
ESM-2 650M	Rep4 site+onehot	XGB	0.8638	0.8958	0.8892	0.8820
ESM-2 650M	Rep1 surprisal + physchem	LR	0.8835	--	--	--
ESM-2 650M	Rep2b site-emb	LR	0.8807	--	--	--
ESM-2 650M	Rep1 surprisal	LR	0.8731	0.8580	0.8608	0.8417
ESM-2 650M	Rep2c Δ site-emb	LR	0.8605	--	--	--
ESM-2 650M	Rep3 surprisal-20	RF	0.8399	0.8458	0.8338	0.8143
ESM-2 650M	Rep2a mean-emb	LR	0.8363	--	--	--
ESM-1v	Rep4 site+onehot	XGB	0.8345	0.8702	0.8748	0.8508
ESM-1v	Rep1 surprisal	LR	0.8357	0.8213	0.8248	0.7973
ESM-1v	Rep3 surprisal-20	RF	0.8114	0.8212	0.8077	0.7895
ESM-1v	Rep2c Δ site-emb	LR	0.8206	--	--	--
ESM-1v	Rep2a mean-emb	LR	0.7964	--	--	--
ESM-1v	Rep2b site-emb	LR	0.7832	--	--	--
ESM C 600M	Rep4 site+onehot	XGB	0.8598	0.8964	0.8991	0.8950
ESM C 600M	Rep2a mean-emb	LR	0.8780	--	--	--
ESM C 600M	Rep1 surprisal	LR	0.8733	0.8608	0.8665	0.8377
ESM C 600M	Rep2b site-emb	LR	0.8643	--	--	--
ESM C 600M	Rep2c Δ site-emb	LR	0.8589	--	--	--
ESM C 600M	Rep3 surprisal-20	RF	0.8373	0.8422	0.8376	0.8163

Reading the table. - **ESM-2 650M and ESM C 600M lead and are close**; ESM-1v trails on every representation, but by an amount that varies by representation and architecture rather than a single band. On surprisal and Rep4 the gap is ~ 0.02 - 0.04 against both architectures. On the raw embeddings the gap is larger and uneven: **ESM C's gap is consistently ~ 0.08** on both Rep2a mean-emb (0.878 vs 0.796, $\Delta 0.082$) and Rep2b site-emb (0.864 vs 0.783, $\Delta 0.081$); **ESM-2's gap is representation-dependent** — small on Rep2a (0.836 vs 0.796, $\Delta 0.040$) but similarly large on Rep2b (0.881 vs 0.783, $\Delta 0.098$). ESM C 600M holds the single best contiguous cell (Rep4, XGBoost, 0.899); ESM-2 650M's best cell (Rep4, XGBoost, 0.889) is a close second. - **Rep4 (per-position surprisal + substitution one-hot) is the strongest representation on the contiguous split** for every architecture — it generalizes to unseen regions better than the raw embeddings. - **Three cells changed classifier under bias-free selection** (Identity floor, Physchem floor, and ESM-2 Rep4 all moved from Random Forest to XGBoost — RF had won those cells narrowly on the contiguous split itself but not on modulo). The changes are small (≤ 0.007 AUROC) and don't affect which architecture or representation is best. - **The random/modulo columns are much higher (0.94-0.96) than contiguous (0.86-0.90), especially for the embedding representations.** This is the generalization story: on random splits the embeddings partly exploit position information that leaks across folds; on the contiguous split — where whole regions are unseen — that advantage disappears and the

compact surprisal+identity representation (Rep4) holds up best. **The contiguous number is the one to trust for a tool that must generalize to new regions.**

3. Is a trained classifier worth its complexity over zero-shot?

The raw masked-marginal surprisal is itself a predictor — no training required. We compared it (as an AUROC ranker) against the best trained classifier, per architecture, using bias-free (modulo-selected) classifiers throughout.

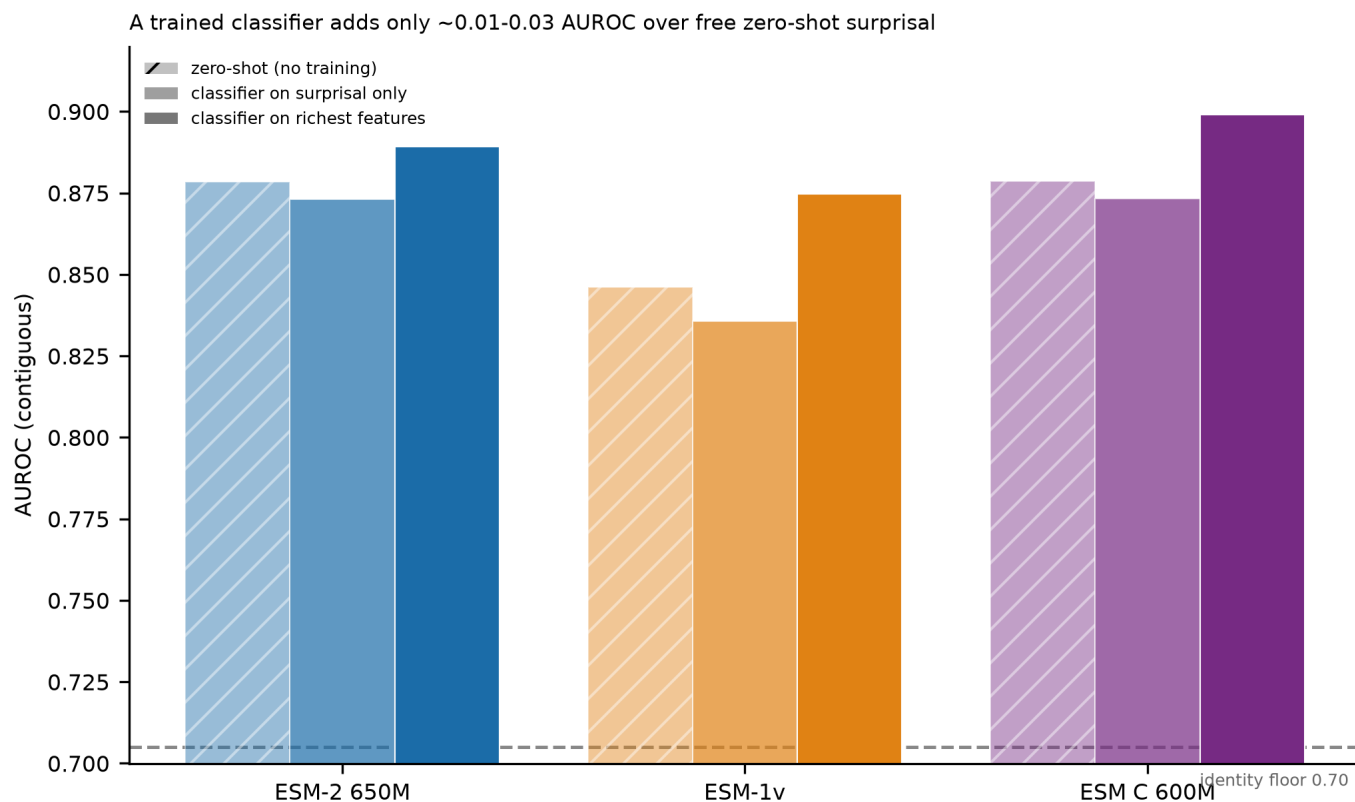


Figure 3. Zero-shot surprisal AUROC vs the best trained classifier, contiguous split. A classifier on surprisal-only adds essentially nothing; only richer features (Rep4) buy a small margin.

Architecture	Zero-shot	Classifier (surprisal only)	Classifier (richest)	Gain over zero-shot
ESM-2 650M	0.8783	0.8731	0.8892	0.0109
ESM-1v	0.8461	0.8357	0.8748	0.0287
ESM C 600M	0.8787	0.8733	0.8991	0.0204

Finding. Zero-shot surprisal alone already reaches **0.85-0.88** on the contiguous split. A classifier trained on surprisal-only does **not** improve on it (within noise — thresholding one scalar cannot beat ranking by it). The gain from a trained classifier comes entirely from *richer features* (Rep4's substitution one-hot on top of surprisal), and even then it is modest: **+0.011 to +0.029 AUROC** (ESM-2's gain shrank slightly under bias-free selection, from +0.017 to +0.011, since its Rep4 cell moved from RF to XGBoost). Practically: if simplicity and deployability matter, zero-shot surprisal captures the large majority of the achievable signal; the trained classifier buys a couple of points, not a transformation.

4. Feature weights: what carries the signal

To rank features without cherry-picking, a single Logistic Regression was fit on **all** features stacked together (2,559 dimensions: surprisal scalars, 20-dim surprisal vectors, site embeddings for ESM-2 and ESM C, physicochemical properties, ESMFold structural features, and identity one-hot), standardized. Coefficient magnitude ranks the signal. This analysis is unaffected by the classifier-selection fix (it fits a single LR on all features jointly, with no per-feature-set classifier choice involved).

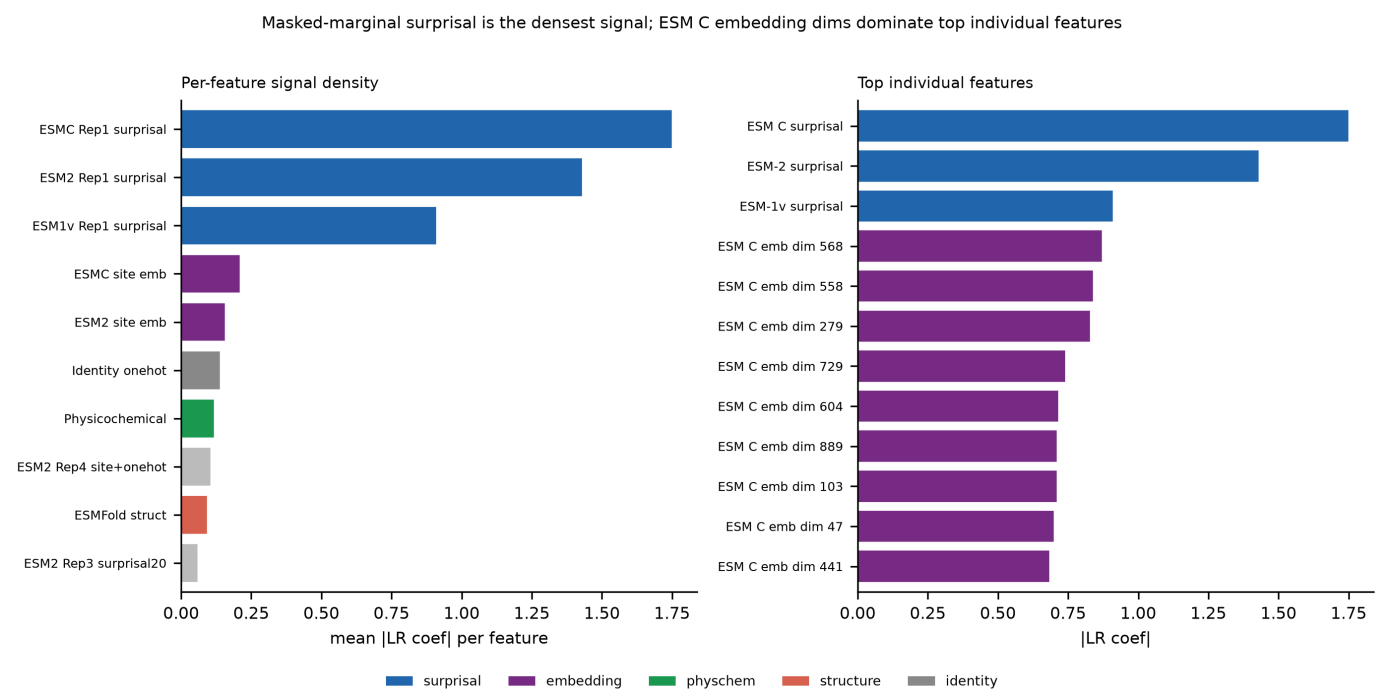


Figure 4. (a) Per-feature signal density (mean |LR coefficient| per feature) by block. (b) Top individual features. The three masked-marginal surprisal scalars are the densest single signals; ESM C embedding dimensions dominate the top individual features.

Feature block	N features	Total weight	Mean weight /feature
ESMC_Rep1_surprisal	1	1.7508	1.7508
ESM2_Rep1_surprisal	1	1.4298	1.4298
ESM1v_Rep1_surprisal	1	0.9101	0.9101
ESMC_site_emb	1152	242.088	0.2101
ESM2_site_emb	1280	202.1359	0.1579
Identity_onehot	40	5.5778	0.1394
Physicochemical	16	1.901	0.1188
ESM2_Rep4_site+onehot	40	4.2146	0.1054
ESMFold_struct	8	0.747	0.0934
ESM2_Rep3_surprisal20	20	1.2134	0.0607

Reading the weights. - The masked-marginal surprisal scalars are the densest signal in the entire matrix — ESM C (1.75), ESM-2 (1.43), ESM-1v (0.91) per feature, the three highest-weighted individual features of anything. This is the project's hypothesis made quantitative: a single surprisal number per variant carries more predictive weight per feature than

any embedding dimension, physicochemical property, or structural feature. - **Site embeddings carry large total weight** (they have >1,100 dimensions each) but ~0.16-0.21 per feature — the signal is real but distributed across many dimensions. - **Physicochemical and identity features contribute modestly** (~0.12-0.14/feature); the physicochemical charge-change term appears among the stronger individual features. - **ESMFold structural features have the lowest total block weight** (0.747 across 8 features; ~0.093/feature) — real but the weakest-contributing block. See §5.

This ranking justifies the feature choices for a final model — surprisal first, site embeddings second — and is derived by giving the linear model every available feature rather than by manual selection.

5. ESMFold predicted-structure arm

A separately-labeled arm tested predicted-structure features. These are **WT position-mapped** structural features (pLDDT, SASA, relative SASA, burial, secondary structure, contact number) from a single ESMFold fold of the wild-type protein, mapped to every variant by position — i.e. *where* a mutation sits structurally, not *which* substitution it is (per-variant delta-fold features, which would be substitution-specific, were not computed here and remain future work). All 4,770 variants are covered, so these AUROCs are directly comparable to the sequence-only arm. Classifier selection uses the same bias-free (modulo-selected) rule as §2.

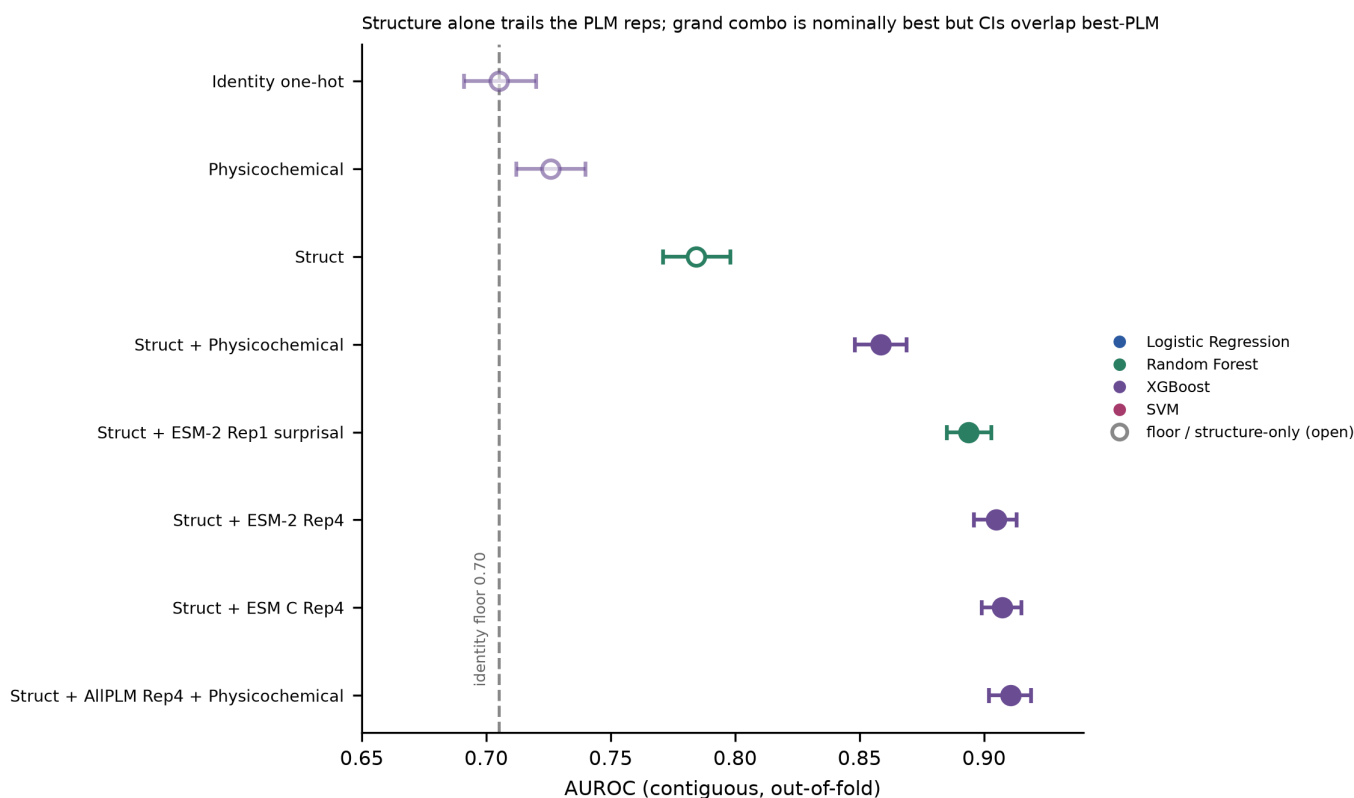


Figure 5. Contiguous AUROC for structure-alone, structure+physicochemical, and structure+PLM combinations. Open markers = floors and structure-alone. The best cell (rightmost) combines structure with all PLM features and physicochemical properties.

Findings. - **Structure alone reaches contiguous AUROC 0.784** (Random Forest, the pre-specified classifier) — above the floors (0.70-0.73) but well below every PLM surprisal representation (0.87-0.90). Position-level structural context carries some signal, consistent with the weight analysis showing it as the weakest block. - **Structure + PLM**

combinations reach ~0.90-0.91. ESMFold + ESM C Rep4 reaches 0.9072; the grand ESMFold + all-PLM Rep4 + physicochemical reaches **0.9107 [0.902,0.919]** (XGBoost) — nominally above the best sequence-only cell (0.8991 [0.890,0.908]), but **the two 95% confidence intervals now overlap** under bias-free classifier selection (grand-combo lower bound 0.902 vs sequence-only upper bound 0.908). An earlier version of this analysis reported these as non-overlapping (+0.018 gain); that used contiguous-biased classifier picks for both cells (Random Forest for the grand combo) and overstated the separation. **The honest read: adding structure to the best PLM features may help a little, but this benchmark cannot distinguish that gain from noise — it is a plausible small effect, not a demonstrated one.**

- **Interpretation.** Predicted structure is a weak feature on its own, and its value in combination with the strongest PLM features is not established at this sample size — it is not ruled out either. Because it is a single WT fold reused across all variants (folded once, not per-variant), the cost of including it in a deployable pipeline would be negligible regardless of whether it turns out to help. Because it is WT position-mapped, it cannot distinguish substitutions at the same position, and it is blind to catalytic effects that do not perturb the fold; its value for substitution-specific or epistatic questions is not tested here. **The final single-mutant model (next session) should run an ablation pass (sequence-only → +physicochemical → +structure → +all) to determine, with adequate power, whether structure contributes anything once the strongest PLM features are already included — rather than assuming either that it does or that it doesn't.**

6. Methods (brief)

- **Data.** Firnberg 2014 DMS for TEM-1 (BLAT_ECOLX), 4770 single-missense variants after sealing 13 wet-lab-panel variants; binary label `DMS_score_bin` (balance 0.50).
 - **Splits.** 5-fold, seed 42. Random (variant KFold), modulo (position mod 5), contiguous (five linear-position blocks 24-76/77-129/130-182/183-234/235-286). Contiguous is primary.
 - **Features.** ESM-2 650M, ESM-1v, ESM C 600M masked-marginal surprisal (Rep1 scalar, Rep3 20-dim vector, Rep4 + substitution one-hot) and mean/site/site-delta embeddings (Rep2a/b/c); physicochemical (16); identity one-hot (40); ESMFold WT position-mapped structural (8).
 - **Classifiers.** Logistic Regression, Random Forest, XGBoost, SVM (RBF), each with inner 3-fold grid search. SVM gated to ≤ 200 -dim sets; tree models gated to ≤ 400 -dim sets on the wide embedding stacks (LR is both faster and stronger there) — a direction-finding scope, not an exhaustive certification.
 - **Classifier selection (bias correction).** The classifier reported for a (feature_set) is the one with the highest **modulo-split** AUROC, applied identically across random, modulo, and contiguous columns. This is a leakage-free choice with respect to the contiguous headline: contiguous scores play no role in picking which classifier's number gets reported on the contiguous split. All four classifiers' contiguous AUROC are additionally reported as a sensitivity block (§2) for full auditability.
 - **Metric.** Out-of-fold AUROC; 95% CIs by 2,000-fold bootstrap; significance by DeLong and 5,000-fold paired bootstrap on paired OOF predictions, using the pre-specified classifier on both arms of the comparison.
 - **Not done (by design).** No model is locked or saved; the 13 sealed wet-lab variants are not scored; per-variant ESMFold delta features were not computed; RF/XGBoost on the wide embedding stacks and distilled-feature model selection are deferred.
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7. Limitations

- **Single dataset, single protein.** One DMS study (Firnberg, TEM-1). External validation on a second β -lactamase or DMS dataset is required before the finding generalizes.

- **Single mutants only.** This benchmark establishes signal for single missense variants; it does not test prediction of unseen double/triple mutants.
- **Direction-finding scope.** The classifier grid was gated for speed (LR-only on wide embeddings, trees deferred there); a final model-selection study would run the full grid on the shortlisted feature sets.
- **Predicted structure is WT-only,** and its incremental value on top of the best PLM features is not resolved by this benchmark (§5) — the point estimate is positive but the confidence intervals overlap, so a dedicated, adequately powered comparison is needed before treating structure as either helpful or not.

Tables (CSV) and figures (PNG) accompanying this document are in [results/](#).